Class 18: Online Learning and Concept Drift

Data-driven Systems Engineering

Agenda

- Why a validated model can become obsolete.
- Offline (batch) versus online (incremental) learning.
- River: learning one observation at a time, with prequential evaluation.
- Concept drift: definition, types, and detection with ADWIN.
- Monitoring a streaming model with River and Evidently.

Offline versus online learning

Offline (batch)

- Trained once on a fixed dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$.
- Straightforward benchmarking on a held-out test set.
- Adapts only through periodic retraining.

Online (incremental)

- Updated after each observation $(\mathbf{x}_t, y_t)$ of a stream, with constant memory.
- Operates directly on unbounded data streams.
- Adapts continuously to change.

- **Prequential evaluation** (test-then-train): each observation is first used to test the model, $\hat{y}_t = f_{t-1}(\mathbf{x}_t)$, and only then to update it, so every prediction is made on unseen data.

Why adaptation becomes necessary

- User behaviour changes.
- Sensors degrade or are recalibrated.
- Business policies or regulations change.
- New error types appear only after deployment.

Key point

Adaptation does not indicate that the initial model was wrong; it acknowledges that the data-generating process evolves over time.

River example: incremental regression

```
model = preprocessing.StandardScaler() | linear_model.LinearRegression(
    optimizer=optim.SGD(0.01)
)
mae = metrics.MAE()

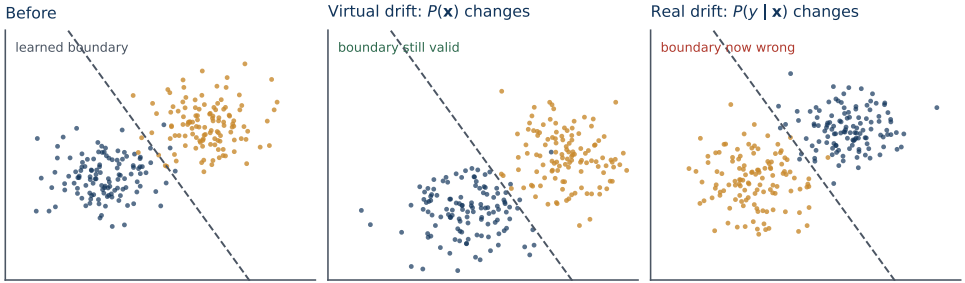
y_pred = model.predict_one(x)
model.learn_one(x, y)
mae.update(y, y_pred)
```

- From `notebooks/Class8_Model_Incremental_learning.ipynb`: the model predicts the current temperature in Paris from the wind speed (`x = {"wind": wind}`), obtained from the Open-Meteo API, which refreshes these values every 15 minutes; polling faster only repeats the same observation.
- `predict_one` precedes `learn_one` (prequential evaluation); the mean absolute error (MAE) and the parameters, by stochastic gradient descent (SGD), are updated at every observation.

What is concept drift?

- Concept drift is a change over time of the joint distribution, $P_t(\mathbf{x}, y) \neq P_{t+\Delta}(\mathbf{x}, y)$; it is a change of the data-generating process, not noise in individual readings.
- Since $P(\mathbf{x}, y) = P(\mathbf{x})\, P(y \mid \mathbf{x})$, two cases are distinguished:
  - **virtual drift** (covariate shift): $P(\mathbf{x})$ changes, $P(y \mid \mathbf{x})$ does not;
  - **real drift**: $P(y \mid \mathbf{x})$ changes, so the learned decision function becomes incorrect.
- A model trained before a real drift may still produce confident predictions that are systematically wrong.

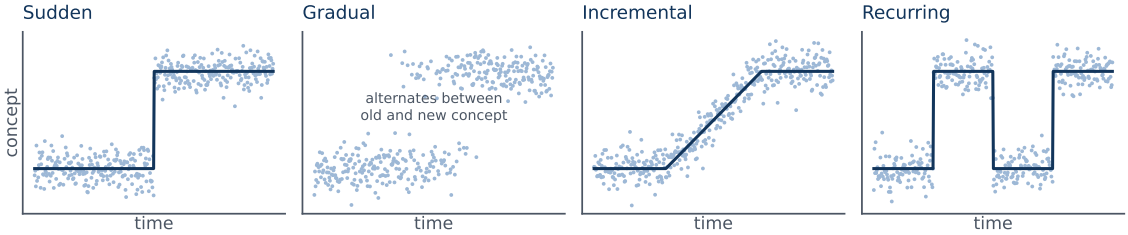
Virtual versus real drift



- **Virtual drift:** the inputs move to a different region of $\mathcal{X}$, but the class of each point is unchanged; the learned boundary remains correct, although the model may now operate where it saw little training data.
- **Real drift:** the same inputs now belong to the other class; the boundary itself is wrong and the model must be updated.
- Monitoring the inputs alone (e.g. a data drift report) detects virtual drift; detecting real drift requires the labels, i.e. the error stream.

Illustrative synthetic data.

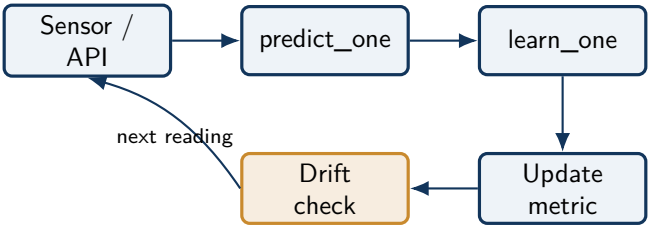
Temporal patterns of drift



- **Sudden:** the concept is replaced at once, e.g. a sensor recalibration or a change of product.
- **Gradual:** samples from the old and the new concept alternate, with the new one becoming more frequent.
- **Incremental:** the concept moves continuously through intermediate states, e.g. wear of heating equipment.
- **Recurring:** previous concepts return, e.g. seasonal patterns; a model can reuse what it learned before.

Illustrative synthetic data; dots are observations, lines the underlying concept.

The streaming learning loop



River example: ADWIN drift detection

```
adwin = drift.ADWIN(delta=0.002)
...
error = abs(y - (y_pred or 0))
adwin.update(error)

if adwin.drift_detected:
    print("Resetting model due to detected drift...")
    model = preprocessing.StandardScaler() | linear_model.LinearRegression(
        optimizer=optim.SGD(0.01)
    )
    adwin = drift.ADWIN(delta=0.002)
```

- From `notebooks/Class8_Model_Incremental_learning.ipynb`: the adaptive windowing detector (ADWIN) monitors the absolute errors and signals a significant change; the notebook then resets both the model and the detector, a strategy examined experimentally below.

How ADWIN works

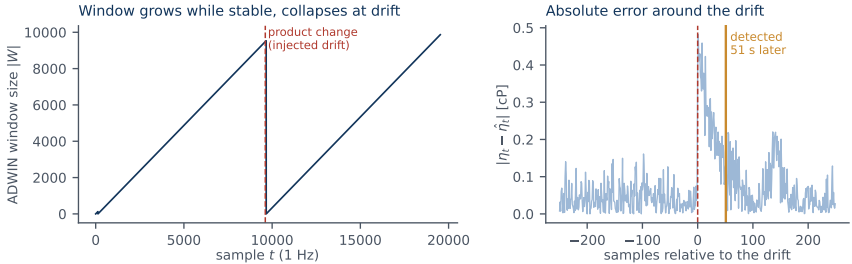
- ADWIN (Bifet and Gavaldà, 2007) maintains a window W of the most recent values (here, absolute errors) and examines every split of W into an older part W_0 and a recent part W_1, of sizes n_0 and n_1.
- A change is detected, and W_0 is dropped, when the two means differ by more than a threshold derived from the Hoeffding bound:

$$|\hat{\mu}_{W_0} - \hat{\mu}_{W_1}| \geq \varepsilon_{\text{cut}}, \qquad \varepsilon_{\text{cut}} = \sqrt{\frac{1}{2m} \ln \frac{4}{\delta'}}, \quad m = \frac{1}{1/n_0 + 1/n_1}, \quad \delta' = \frac{\delta}{n}.$$

- The confidence parameter δ (here 0.002) controls the false-alarm rate: the window grows while the stream is stable and shrinks as soon as its mean changes.
- This provides a statistically grounded trigger for retraining or resetting, instead of a fixed retraining schedule.

ADWIN on the pasteurization stream

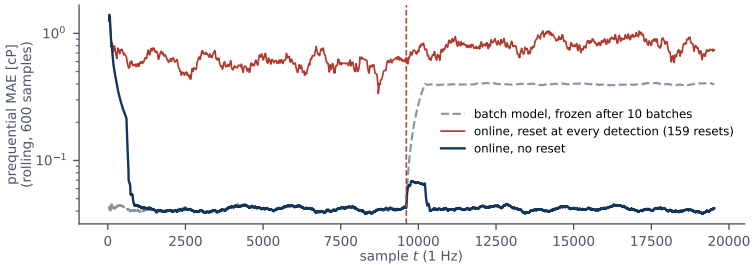
Experiment: an online soft sensor estimates the viscosity η from the temperature T (60 batches, 19,530 samples at 1 Hz). After batch 30, a product change raises η by 0.4 cP at every temperature: a sudden real drift.



- The ADWIN window grows linearly while the error stream is stationary and collapses when the drift is detected, 51 samples (51 s) after it occurred.
- Apart from one alarm during the initial learning phase, no false alarm is raised in 19,530 samples ($\delta = 0.002$).

Data from `pasteurization/synth_sensors.py` with an injected drift; River 0.22, `LinearRegression` with SGD, `ADWIN(delta=0.002)`.

Adaptation strategies compared



Strategy	MAE before	MAE after	
Batch model, frozen after 10 batches	0.041	0.398	does not adapt
Online, no reset	0.052	0.043	recovers in about 200 samples (3.5 min)
Online, reset at every detection	0.599	0.820	159 resets: reset loop

- A reset replaces the model by an untrained one, whose large and decreasing error is itself a change that ADWIN detects: the model never stabilizes. Safer responses are to warm-start from the current model, retrain on the recent window W_1, or only raise an alert.

Monitoring beyond a single detector

- `notebooks/Class8_Model_Incremental_learning.ipynb` also produces Evidently reports (`DataDriftPreset`, `DataQualityPreset`) alongside River.
- ADWIN on the error stream is a fast, targeted signal that detects degradation of performance; a drift report tests the distribution of every input column and can reveal virtual drift before performance degrades.
- Combining a lightweight online detector with periodic, more comprehensive reports is a common production pattern.

Batch retraining versus online learning

Latency of adaptation	Online learning adapts immediately; batch retraining waits for the next cycle.
Auditability	Batch models are fixed snapshots that can be reviewed and versioned; the state of an online model changes continuously.
Stability	Online models may overreact to noise if the detector or the learning rate is too sensitive.
Infrastructure	Online learning avoids large periodic training jobs but requires an always-on service.

Drift in the two running cases

- **Fraud:** seasonal spending patterns and new merchant categories (virtual drift); fraudsters adopting new tactics (real drift).
- **Pasteurization:** sensor recalibration (sudden), raw-material variability, or wear of heating and cooling equipment (gradual).
- In both cases, drift detection turns "the model has become worse" from an unexplained observation into a specific, timestamped event.

Repository connections

Repo activity

- `notebooks/Class8_Model_Incremental_learning.ipynb`: River regression, ADWIN, and Evidently reports.
- `streamlit/pages/App_Stream.py`: the same River and ADWIN pattern as a live dashboard, plotting detected change points on the MAE curve.

Papers and materials

Papers and materials

- Gama, J., Žliobaitė, I., Bifet, A., Pechenizkiy, M., Bouchachia, A. (2014). A survey on concept drift adaptation. *ACM Computing Surveys*, 46(4), 44:1–44:37.
- Bifet, A., Gavaldà, R. (2007). Learning from time-changing data with adaptive windowing. *Proceedings of the SIAM International Conference on Data Mining*, 443–448.
- Lu, J., Liu, A., Dong, F., Gu, F., Gama, J., Zhang, G. (2019). Learning under concept drift: A review. *IEEE Transactions on Knowledge and Data Engineering*, 31(12), 2346–2363.
- Montiel, J., et al. (2021). River: machine learning for streaming data in Python. *Journal of Machine Learning Research*, 22(110), 1–8.
- Gama, J., Sebastião, R., Rodrigues, P. P. (2013). On evaluating stream learning algorithms. *Machine Learning*, 90(3), 317–346.
- Documentation: River; Evidently.

Take-away message

The data-generating process changes over time:
evaluate models prequentially, detect drift
with a statistical test, and define in advance
how the model is updated when drift is detected.

Thank you for your attention.

Next: Class 19 — hands-on lab: streaming the pasteurization sensors.